

Introduction to PL/SQL

Gate Smashers Series | Core Database Concepts & Programming Extensions

1. Relevance & Target Examinations

While standard SQL is the primary focus for competitive exams like **GATE**, basic to advanced PL/SQL is essential for several other key technical exams and academic curricula:

- **Recruitment Exams:** KVS (Kendriya Vidyalaya Sangathan), DSSSB, and state-level IT officer exams.
- **Industry Certifications:** Oracle Database SQL / PL/SQL Certification track.
- **Academic Syllabi:** Class 11th & 12th Computer Science / Informatics Practices curricula.

2. What is PL/SQL?

PL/SQL stands for **Procedural Language extension to Structured Query Language**. Standard SQL is non-procedural/declarative, whereas PL/SQL introduces full programming capabilities into standard database queries.

Feature	Standard SQL	PL/SQL
Nature	Declarative ("What to do")	Declarative + Procedural ("What & How to do")
Execution Unit	Single queries (1–2 lines)	Entire block / program code
Capabilities	Data retrieval & manipulation	Variables, functions, loops, error handling
Query Style	WRITE A QUERY TO...	WRITE A PROGRAM / CODE TO...

3. Key Components of PL/SQL

Procedures & Functions

Reusable subprograms with defined parameters and logic execution (most asked exam topics).

Triggers

Stored programs automatically invoked in response to specific database table events.

Cursors

Pointers to context areas allowed to process individual rows of multi-row query results.

Control Structures

Full support for conditionals (IF-THEN) and loops (FOR, WHILE) similar to C language.

Key Concept Note: In standard C programming, computing logic requires function declaration, argument passing, and math operations (e.g., $c = a + b$). PL/SQL enables this identical programmatic logic directly inside database engines.

4. Architecture of a PL/SQL Block

Every PL/SQL block consists of three major sections. Understanding this standard block structure is fundamental to writing correct procedural SQL code.

1. Declaration Section (Optional)

Defines variables, constants, cursors, and user-defined exceptions. Note: Variable name comes *before* the datatype.

2. Executable Section (Mandatory)

Contains procedural statements enclosed within BEGIN and END ; keywords.

3. Exception Handling Section (Optional)

Defines manual or system error-handling logic (e.g., handling ZERO_DIVIDE runtime errors).

5. Syntax Rules & Assignment Operators

- **Variable Declaration Syntax:** `variable_name datatype;` (e.g., `a INT;`, `b NUMBER;`)
- **Assignment Operator (`:=`):** Used to assign or pass values to a variable (e.g., `c := a + b;`).
- **Comparison Operator (`=`):** Used strictly to check equality between two expressions.

6. Basic Code Structure & Example

Below is the foundational PL/SQL program structure for performing addition of two numbers:

```
-- 1. Declaration Section DECLARE a INT; b INT; c INT; -- 2. Executable Block BEGIN a :=
```

Next Steps in the Series: In subsequent sessions, detailed syntax, execution steps, and practical questions on **Procedures** and **Triggers** will be covered.